QUINT Protocol - Expanded Details

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Folder & File Labeling

1. Create a New Folder
    a. Label this “Quint_Workflow”
2. Within this folder create seven sub-folders
    a. 1. Nutil_TiffCreator_Input
    b. 2. Nutil_TiffCreator_Output
    c. 3. Nutil_Transform_Output
    d. 4. Nutil_Resize-QuickNii_XML_JSON
    e. 5. VisuAlign
    f. 6. Ilastik
    g. 7. Nutil_Output
3. Put your desired images into your first subfolder (1. Nutil_TiffCreator_Input)
4. Relabel each file to have ‘_sXXX’ at the end of each name (DO NOT FORGET THIS STEP!!)
    a. Do not reuse the same three digit number
        i. I.e. 001 should not be at the end of more than one file

Parsed
KEK95 _mid2VP_10x_7x5_250109_s003_thumbnail.png
KEK95_antVP_10x_7x5_250109_s001_thumbnail.png
KEK95_mid1VP_10x_7x5_250109_s002_thumbnail.png
KEK95_postVP_10x_7x5_250109_004_thumbnail.png

    b. Additionally, put an underscore for every space in the file name
        i. I.e. “KEK99_antVP_10x_7x5_250109_s001”

Nutil Part 1

1. Download [Nutil](https://www.nitrc.org/projects/nutil/) (https://www.nitrc.org/projects/nutil/)
    a. Need to extract files in order to open programs - extract all to desktop, then you must open from the programs subfolder (lizard icon once extracted) (similar to fiji)

nutil	2/6/2025 11:37
nutil_program	2/6/2025 11:36
opengl32sw.dll	2/6/2025 11:36

2. Open Nutil
3. Go to ‘Operations’ Tab
4. Click New
    a. In dropdown choose ‘TiffCreator’
    b. Make a name for your project

        i. I.e. KEK99_TiffCreator
    c. Choose lzw for ‘Output compression’
    d. For ‘Input folder’ choose your folder full of images
        i. Should be labeled ***“1. Nutil_TiffCreator_Input”***
    e. For ‘Output folder’ choose your second folder
        i. Should be labeled ***“2. Nutil_TiffCreator_Output”***
    f. *Keep tile size 128*
    g. Click the ‘Start’ button in bottom left and it will ask to save a text file
        i. Make name specific, i.e. “TiffCreator”
    h. After you start the program should automatically change to ‘Process output’ tab
        i. Depending on the amount of images chosen, the task may take a minute or may be already completed
    i. The files should now be found in the output folder chosen
5. Go back to the ‘Operations’ tab
    a. Click ‘New’ again and in dropdown choose ‘Transform’
    b. Again make a name for your project
        i. I.e. KEK99_Transform
    c. Keep ‘Output compression’ lzw
    d. For ‘Input folder’ choose your previous output folder
        i. I.e. ***“2. Nutil_TiffCreator_Output”***
    e. For ‘Output folder’ choose your third folder
        i. Should be labeled ***“3. Nutil_Transform_Output”***
    f. In the dropdown next to *‘Auto cropping’ click “No”*
    g. Because auto cropping is off you DON’T need to change anything for ‘Image background color’ or ‘Background color range’
        i. They should both be blank automatically
    h. In the box next to ‘List of images’ click ‘Fill files’
        i. This should put all the files from your input folder into the box
    i. Click ‘Show advanced settings’
    j. In the dropbox next to *‘Only create thumbnails’ choose “Yes”*
    k. In the dropbox next to ‘Thumbnail size’ choose desired number
        i. If unsure go with 0.5
    l. Click the ‘Start’ button and it will again ask to save a text file
        i. Name it and click save
    m. After clicking save the program should again switch over to the ‘Process output’ tab
        i. Depending on the amount of images chosen, the task may take a minute or may be already completed
    n. The files should now be saved into the chosen output folder in a subfolder labeled ‘thumbnails’

6. Go back to 'Operations' tab
    a. Click 'New' again and in dropdown choose 'Resize'
    b. Create a name for your project
    c. For 'Input folder' choose your previous transform output folder
        i. Should be labeled ***"3. Nutil_Transform_Output"***
    d. For 'Output folder' choose your fourth folder
        i. Should be labeled ***"4. Nutil_Resize-QuickNii_XML_JSON"***
    e. 'Resize type' will be kept as "Percent"
    f. Next to 'Resize factor' choose your desired percentage
        i. The smaller the percentage the more pixelated the image will in QuickNii, if *unsure of a percentage choose "80"*
    g. Click the 'Start' button again and save your text file
        i. Name it and click save
    h. After clicking save the program should again switch over to the 'Process output' tab
        i. Depending on the amount of images chosen, the task may take a minute or may be already completed
    i. The files should now be saved into the chosen output folder

QuickNii

7. Download QuickNii (**https://www.nitrc.org/projects/quicknii**)
    - **Note that working with the Calabrese rat P80 atlas (https://scalablebrainatlas.incf.org/rat/CBWJ13_age_P80) is not readily available on the QuickNII download website. You must download the "QuickNII-Calabrese_P80_Rat-Test.zip" file from this address (https://www.nesys.uio.no/P80_Rat/). If you are interested in using other rat or mouse atlases, see the dropdown menu at the QuickNII download website.**
8. Minimize the *Nutil* program and go to File Explorer
    a. In File Explorer go to downloads and click the extracted QuickNii folder - when downloading, ensure you are choosing the one with the correct atlas (rat!)
    b. Within the folder there should be a file named "FileBuilder"
    c. Open the program
    d. In the program find your resize output folder and choose desired images
        i. They may be in another folder within your output folder named "Thumbnails"
    e. Click 'Open' and your files should appear in the table
    f. Under 'Remark' there should be a green 'OK'
        i. If there is not it will say "Name mismatch" – this means that you did not correctly number your images using the "_sXXX" format and that two files have the same three digits – go back and ensure that everything is labeled correctly

g. At the bottom of the screen click ‘Save XML’
h. It will ask you again to name your file
    i. You can choose the same name you used when entering the files in, i.e. “Resize 80 (or chosen percentage)” or “Images 87-102” to help you keep track of each XML file
    ii. **Make sure to change where the file will be saved in, it automatically will choose ‘Documents’ – when saved in documents the images will not open in QuickNii!

→ **PUT THE XML IN THE SAME FOLDER AS YOUR RESIZE OUTPUT: images will not open in Quicknii unless they are in the same folder**

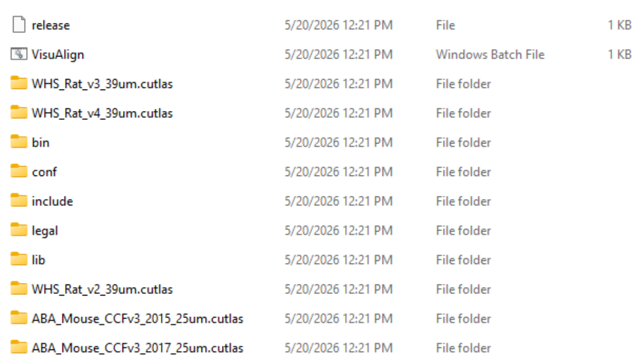
i. The XML file (only one file for all images) will be saved as Microsoft Edge HTML Document

9. Close the Builder and open up the actual QuickNii program
    a. On the top bar choose the ‘Manage data’ tab
    b. This will open up a table and chart
    c. Below the table click the ‘Load’ button
    d. Find your XML files and click ‘Open’
        i. Should be in your fourth folder ***“4. Nutil_Resize-QuickNii_XML_JSON”***
    e. All files will show up in the table
    f. Double click desired image
        i. The number that was used to label them at the beginning should be under the ‘s#’ column – this will help to identify what image you are looking at
    g. The image should then appear on top of the atlas
    h. To find the exact part of brain use the three options on the right side (Sagittal, Coronal & Horizontal) – you can click anywhere and rotate the atlas with these options
    i. In order to make the atlas taller or wider use the arrow buttons right above the atlas/image
        i. There should be 9 options
        ii. The last three are what you should focus on
    j. If you click either the up/down arrow or the left/right arrow the arrow should appear on the image
        i. Use the spacebar to move the arrow around, after you release the space bar the arrow should stay in place
        ii. Now on the opposite side from the arrow you can drag the atlas around
            When you release the mouse the pointer will become the four way arrow and not be an up/down or right/left arrow
            You will need to re-click those arrows in order to move it up/down or right/left

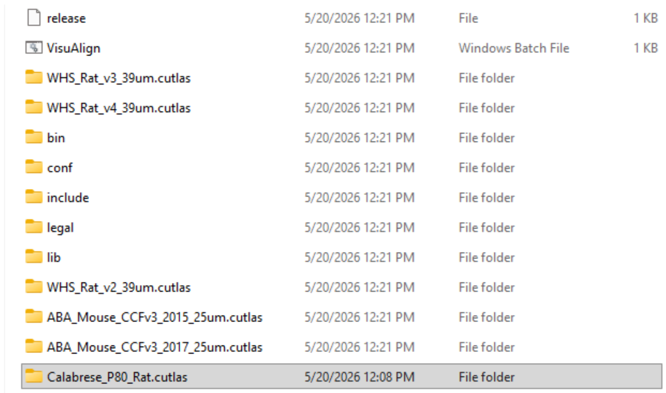
k. After you are happy with your atlas placement click the ‘Store’ button above the arrow buttons
    i. A green exclamation mark will appear where the red question mark was previously
l. You can now go to another image by clicking the right or left arrow next to the green exclamation mark
m. After you are done with all your images, click on the ‘Manage Data’ button on the top bar
    i. Should be a little left of the green exclamation mark
n. Your data table and graph should appear and below the data table you will click the ‘Save JSON’ button
o. Save this into the same folder as your XML
    i. Should be labeled ***“4. Nutil_Resize-QuickNii_XML_JSON”***
    ii. Create a specific name to remember it i.e. *“QuickNii_JSON”*
p. Additionally under the table click the “Export Slices” button and save it in the same folder

VisuAlign

10. Download [VisuAlign](https://www.nitrc.org/projects/visualign) (https://www.nitrc.org/projects/visualign)
    a. Need to extract files in order to open program - extract all to desktop, then you must open from the programs subfolder
    b. **Note that working with the Calabrese rat P80 atlas (https://scalablebrainatlas.incf.org/rat/CBWJ13_age_P80) is also not readily available when downloading VisuAlign from the website. You must download the “VisuAlign_cutlas_folder” file from this address (https://www.nesys.uio.no/P80_Rat/) (or from our github repository), extract, and manually move the contained “Calabrese_P80_Rat.cutlas” folder into the extracted Nutil folder. Only WHS Rat and ABA Mouse atlases are available natively.**



→ Add extracted Calabrese cutlas folder…



11. Open VisuAlign
12. Once it is open select the ‘File’ dropdown
    a. Located on the top left
13. Select ‘Open’ and choose your JSON file that you just saved from Quicknii
    a. You image should then appear in the application
14. Drag the ‘Opacity’ slider all the way to the right
    a. Towards ‘Outline”
15. To move the outline to fit your image you will need to move your mouse to the exact place you want to drag and click your spacebar – a + will appear
    a. DO NOT hold the spacebar as that will create many +
16. You can now move the + however you want
    a. You can also always go back to a + and move it again
17. Move the outlines to fit your image
18. Once happy, go to the ‘File’ dropdown again and this time click “Save As…” and save your file into a folder – should save as a JSON
    a. Should go into your fifth folder labeled ***“5. VisuAlign”***
    b. Name the file something specific to decipher between the JSON files
        i. I.e. *“VisuAlign_JSON”*
19. Then go back to VisuAlign and in the ‘File’ dropdown click ‘Export…’ and save to the same folder as your ‘Save As…” files
20. Open ImageJ
    a. There will be two PNG exported from VisuAlign – you will want to focus on the one that ends with ‘_nl’ only - just delete the ‘_nl_rbw’ files to get them out of the way
    b. Keep the .flat files
21. You can use the VisuAlign to Glasbey [Macros Script]instead of manually dropping each image in; otherwise
    a. Put one image that ends with ‘_nl’ into ImageJ
        i. Go to ‘Image’ dropdown and change ‘Type’ to 32-bit
        ii. Then go again to ‘Image’ dropdown and under ‘Lookup Tables’ choose ‘glasbey on dark’

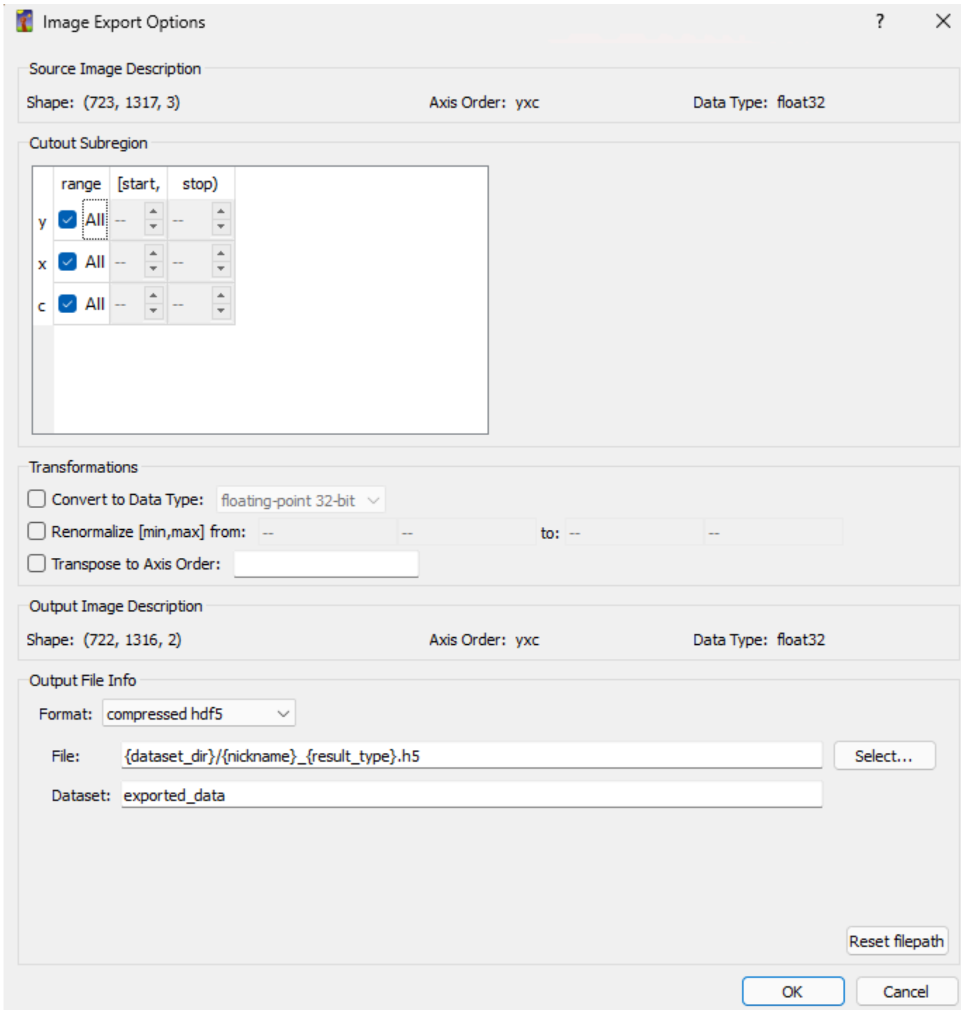
            1. Go back to ‘Image’ dropdown change it back to ‘RGB color’
      iii. Under ‘File’ choose ‘Save As’ and choose “PNG…” and save your new image to the same folder
            1. Labeled ***“5. VisuAlign”***
            2. To differentiate between files rename or add ‘_glasbey’ to the end
   b. Repeat a-d for all images ending in ‘_nl’

NOTE: Resulting images will not be scaled properly. To undertake downstream processing, the QUINT based workflow allows for use in Ilastik and Nutil once again to develop a segmented output. In our lab, we chose to pivot back to FIJI to incorporate the registered atlas images into a pipeline incorporating ROIs from hand-selected cells (see FIJI_scripts folder for method).

Ilastik - downstream quantification processing (not covered in tutorial videos)

22. Download [Ilastik]
    a. Will ask for a PIN in order to download
23. Open Ilastik
    a. The folder was not found in downloads so we had to search in File Explorer for ‘Ilastik’ – it was the second option **below** ‘uninstall ilastik’
24. In the application under ‘Create New Project’ select ‘Pixel Classification’
    a. Save you project into the same folder as your VisuAlign JSON files
    b. If the program automatically opens a project for you, go to the left upper corner and in the ‘Project’ dropdown click ‘Close’ – this should bring you to the main page
25. The application should open a new screen and you will be in the ‘Input Data’ section
    a. You can see all the sections on the left sidebar
26. In the middle of the screen there should be three tabs: ‘Raw Data’, ‘Prediction Mask’ and ‘Summary’
27. Under ‘Raw Data’ there will be a dropdown menu called ‘+Add New…’
    a. You will then click ‘Add Separate Image(s)...’
28. This will open your File Explorer and you will choose your PNG images from your fourth folder
    a. ***“4. Nutil_Resize-QuickNii_XML_JSON”***
    b. The PNG that you add should only end with “_thumbnail”
        i. Not “DTI”, “MRI” or “Segmentation”
29. After your images have been uploaded go to the next section on the left sidebar ‘Feature Selection’ and click ‘Select Features…’
    a. You will want to select all options and press ‘OK’
        i. Will give you a green check mark when selected
30. Then you will go to the third section on the left side ‘Training’

31. There will be two Label buttons named ‘Label 1’ and ‘Label 2’ – rename one to be “Background” and the other “Cells”
32. You can now zoom in (using CTRL and your mouse dial) on your cells and color in some of the cells
    a. Make sure you’re under the “Cells” label and not “Background”
33. Switch to your “Background” label and color in anything that is not a cell
    a. You do not have to color in the whole background – try and choose a wide variety of background colors
34. You can check the coloring by clicking ‘Live Update’ under the label selection
    a. This should update your image to show your cells in yellow
    b. In the bottom left corner a ‘Group Visibility’ box will appear and you can toggle the settings to see your image with different overlays
    c. If some of the background is yellow, then color in those sections with your “background” label
35. Once you’re happy with your cells go to the ‘Prediction Export’ section and there should be a dropdown menu next to ‘Source’
    a. Choose ‘Probabilities’
    b. Under that dropdown there should be a button labeled ‘Choose Export Image Settings…’ click into that and ensure that the format is ‘compressed hdf5’
        i. *You might need to ‘Convert to Data Type’ but we don't know which one*



   ii. Click 'OK'

c. Then you will 'Export All'

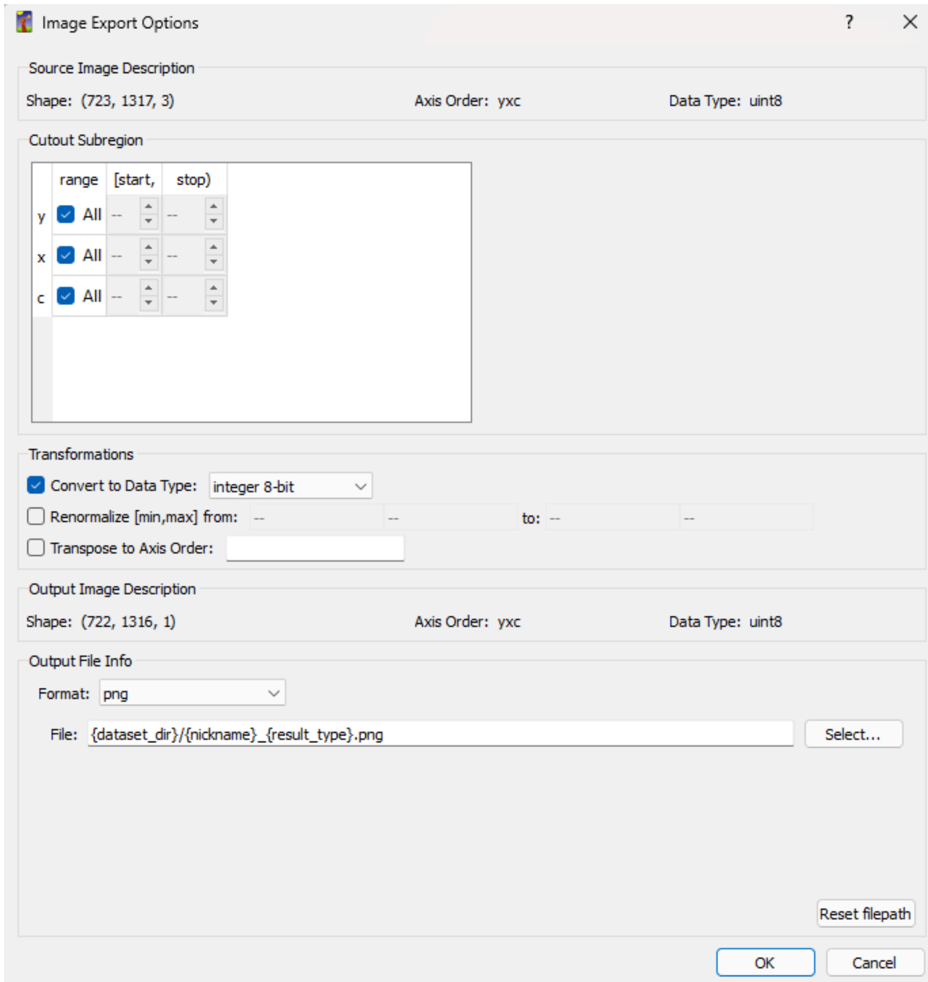
d.

36. After that is exported you will change your source in the dropdown menu from 'Probabilities' to 'Simple Segmentation'

   a. Go back into 'Choose Export Image Settings…' and change format to 'png'

      i. Also change 'Convert to Data Type' to 'integer 8-bit'

         - even though ours is already uint8, it will automatically change it to float32. D/A said we need to convert 'unsigned 8-bit', but that option doesn't appear in our 'Convert to Data Type' drop-down. However, when we select integer 8-bit, the output Data Type preview is called 'uint8', which means unsigned 8-bit, thus it appears that selecting integer 8-bit is already considered an unsigned 8-bit in the program (perhaps because it is a new version of ilastik?).



ii. Click 'OK'

b. Then 'Export All'

37. Open ImageJ
    a. There should be two different file types within your folder from *Ilastik* – PNG and H5 files
    b. You will focus on the PNG files
38. You can use the Ilastik to Glasbey Macros Script instead of manually dropping each image in; otherwise
    a. Open one PNG file at a time in ImageJ
        i. ** Don't freak out, the images should be black
        ii. Go to the 'Image' dropdown and in 'Lookup Tables' choose 'glasbey' - *not 'glasbey on dark!'*
        iii. Then go to 'File' dropdown and choose 'Save As'
            1. To differentiate between files rename or add '_glasbey' to the end
    b. Repeat steps a-c for each 'Simple Segmentation' image

Nutil Pt. 2 - downstream quantification processing (not covered in tutorial videos)

39. Open Nutil
40. Go to ‘Operations’ Tab
41. Click New
    a. This time in the dropdown choose ‘Quantifier’
    b. Label your project
    c. Next to ‘Segmentation folder’ add your folder from *Ilastik*
    d. Next to ‘Brain atlas map folder’ add your folder from *VisuAlign*
    e. Choose your ‘Reference atlas’
        i. i.e. ‘WHS Atlas Rat v4’
    f. Next to ‘XML or JSON anchoring file’ choose your JSON file from VisuAlign
        i. There should be two JSON files one from *QuickNii* and the other *VisuAlign*
    g. Choose your seventh folder as your ‘Output folder’
        i. Should be labeled ***“7. Nutil_Output”***
    h. Choose your ‘Object color’ to match the color of your cells
        i. i.e. red if you’ve done all of the steps correctly
    i. Change ‘Object Splitting’
        i. To ‘No’ if you’re looked at cells
        ii. To ‘Yes’ if you’re looking at something larger
    j. Then click ‘Start’
    k. Your images should then process and load into your output folder
42. Go to File Explorer and open your output folder
    a. Go to ‘Reports’ folder
    b. Open ‘_Objects’ folder
    c. Open ‘_Objects_All’ in notepad and in the first line add “sep=;” and enter so that is the only thing in the first line
    d. Go to ‘File’ and ‘Save’
    e. Now open ‘_Objects_All’ in Excel

Section ID	Object pixe	Object are	units	Center X	Center Y	Region ID	Region Na	Custom Region
_s003	98	98	pixels	15.2551	395.9286	82	Basal forebrain region, unspecified	
_s003	50	50	pixels	11.1	380.3	82	Basal forebrain region, unspecified	
_s003	49	49	pixels	31.55102	401.5714	82	Basal forebrain region, unspecified	
_s003	49	49	pixels	180.8776	420.9184	82	Basal forebrain region, unspecified	
_s003	40	40	pixels	244.075	183.025	82	Basal forebrain region, unspecified	
_s003	40	40	pixels	1001.375	357.6	82	Basal forebrain region, unspecified	
_s003	39	39	pixels	1104.333	302.4359	82	Basal forebrain region, unspecified	
_s003	33	33	pixels	84.39394	455.0909	82	Basal forebrain region, unspecified	
_s003	32	32	pixels	31.625	383.1563	82	Basal forebrain region, unspecified	
_s003	32	32	pixels	55.96875	387.6875	82	Basal forebrain region, unspecified	
_s003	31	31	pixels	113.9677	398.9355	82	Basal forebrain region, unspecified	
_s003	30	30	pixels	246.3667	413.3667	82	Basal forebrain region, unspecified	
_s003	28	28	pixels	1036.893	297.8214	82	Basal forebrain region, unspecified	
_s003	26	26	pixels	40.65385	417.6538	82	Basal forebrain region, unspecified	
_s003	22	22	pixels	956.2273	163.8182	193	Ventral pallidum	
_s003	22	22	pixels	994	214	193	Ventral pallidum	
_s003	16	16	pixels	214.5	378	193	Ventral pallidum	
_s003	16	16	pixels	1094.438	295.5625	82	Basal forebrain region, unspecified	
_s003	14	14	pixels	1062.786	320.3571	82	Basal forebrain region, unspecified	
_s003	14	14	pixels	1064.5	264.7143	82	Basal forebrain region, unspecified	
_s003	13	13	pixels	1008.615	100.6154	193	Ventral pallidum	
_s003	12	12	pixels	319.25	164.5833	82	Basal forebrain region, unspecified	
_s003	12	12	pixels	161.75	429.1667	82	Basal forebrain region, unspecified	
_s003	12	12	pixels	87.5	461.5	82	Basal forebrain region, unspecified	
_s003	10	10	pixels	926.7	206.3	193	Ventral pallidum	
_s003	9	9	pixels	205.1111	360.3333	193	Ventral pallidum	
_s003	8	8	pixels	223.125	236.125	193	Ventral pallidum	
_s003	8	8	pixels	971.875	334.875	82	Basal forebrain region, unspecified	
_s003	8	8	pixels	1105.625	325.25	82	Basal forebrain region, unspecified	
_s003	6	6	pixels	269.8333	389.8333	82	Basal forebrain region, unspecified	
_s003	5	5	pixels	172.2	425.6	82	Basal forebrain region, unspecified	
_s003	5	5	pixels	239.4	411.2	82	Basal forebrain region, unspecified	
_s003	5	5	pixels	937.2	216.4	193	Ventral pallidum	
_s003	5	5	pixels	1020.8	298.6	82	Basal forebrain region, unspecified	
_s003	5	5	pixels	1134.8	314.2	82	Basal forebrain region, unspecified	
s003	4	4	pixels	1024.5	299.5	82	Basal forebrain region, unspecified	

Region pixels	Number of pixels representing the region.
Region area	Area representing the region (region pixels x pixel scale).
Area unit	Region area unit (unit of the pixel scale).
Object count	Number of objects located in the region (N/A if object splitting "ON").
Object pixels	Number of pixels representing objects in this region.
Object area	Area representing objects in this region (object pixels x pixel scale).
Load	Ratio of Object pixels to Region pixels (Object pixels / Region pixels).